Safety Data Sheet

1. PRODUCT AND COMPANY INFORMATION

○ **Product name** : 특수혼합분

○ **Synonyms** : Bluemix Powder, HSPM308B, 521B

○ **Company** : Hyundai Steel Company

○ **Address** :

Pangyo Office: 9th floor, GREITS PANGYO, 117 Bundangnaegok-ro, Bundang-gu, Seongnam-si, Gyeonggi-do

Dangjin Steel Mill: 1480 Bukbosaneop-ro, Songak-eup, Dangjin-si, Chungcheongnam-do

○ **Telephone** : Pangyo Office: + 82 31 510 2114 / Dangjin Steel Mill: + 82 41 680 0114

○ **Recommended use of the chemical and restrictions on use**

Recommended use : Metals and Alloys (This product can be used for various purposes. The manufacture of iron and steel; not limited to carbon, magnesium, chromium, nickel and other alloys with elements forming steel.)

Restrictions on use : Use for recommended use only

2. HAZARDS IDENTIFICATION

○ **Globally Harmonized System of Classification and Labelling of Chemicals (GHS)**

Physical hazard :

Pyrophoric solids Cat. 1

Health hazard :

Acute Toxicity (Inhalation) Cat. 2

Skin sensitization Cat. 1

Environment hazard :

Hazardous to the aquatic environment – short-term (acute) aquatic hazard Cat. 1

Hazardous to the aquatic environment – long-term (chronic) aquatic hazard Cat. 2

Label elements including precautionary statements

Symbol : Signal word : Danger

Hazard statements :

H250 Catches fire spontaneously if exposed to air

H330 Fatal if inhaled

H317 May cause an allergic skin reaction

H400 Very toxic to aquatic life

H411 Toxic to aquatic life with long lasting effects

Precautionary statements :

Prevention :

P210 Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.

P222 Do not allow contact with air.

P231+P232 Handle and store contents under inert gas, Protect from moisture.

P233 Keep container tightly closed

P280 Wear protective gloves/protective clothing/eye protection/face protection.

P260 Do not breathe dust/fume

P271 Use only outdoors or in a well-ventilated area.

P284 [In case of inadequate ventilation] wear respiratory protection.

P261 Avoid breathing dust/fume

P272 Contaminated work clothing should not be allowed out of the workplace.

P273 Avoid release to the environment.

Response :

P302+P335+P334 IF ON SKIN: Brush off loose particles from skin. Immerse in cool water[or wrap in wet bandages].

P370+P378 In case of fire: Use water spray, alcohol-resistant foam, powder fire extinguishing agent, carbon dioxide to extinguish.

P304+P340 IF INHALED: Remove person to fresh air and keep comfortable for breathing.

P310 Immediately call a POISON CENTER/doctor.

P320 Specific treatment is urgent

P302+P352 IF ON SKIN: Wash with plenty of water

P333+P313 If skin irritation or rash occurs: Get medical advice/attention.

P321 Specific treatment

P362+ P364 Take off contaminated clothing and wash it before reuse.

P391 Collect spillage.

Storage :

P403+ P233 Store in a well-ventilated place. Keep container tightly closed.

P405 Store locked up.

Disposal :

P501 Dispose of contents/container to accordance with all applicable regulations.

○ NFPA Rating

Health : 4　　Flammability : 4　　Reactivity : 0　　Water reactivity : 0

3. COMPOSITION/INFORMATION ON INGREDIENTS

Ingredients	Common Name	CAS No.	EINECS No.	Conc. %
Iron : Powder	Iron wire; Iron Powder; Carbonyl Iron	7439-89-6	231-096-4	92.0 ~ 98.0 %
Copper	Copper flakes; RANEY COPPER	7440-50-8	231-159-6	2.00 ~ 5.00 %
Carbon	Activated Carbon; Carboxyl Graphene	7440-44-0	231-153-3	0.10 ~ 3.00 %

4. FIRST AID MEASURES

○ In case of eye contact

- In case of contact with the substance, immediately wash skin and eyes with running water for at least 20 minutes.
- Call 911 or emergency medical institutions.
- If irritation persists in your eyes, seek medical treatment and advice.

○ In case of skin contact

- Wash thoroughly with soap and water.
- Remove contaminated clothing and shoes and isolate them.
- Wash clothes and shoes thoroughly before reuse.
- If you feel uncomfortable, seek medical attention/advice.
- Remove contaminated clothes and shoes and isolate the contaminated area.
- In case of contact with the substance, immediately wash skin and eyes with running water for at least 20 minutes.

- Wash contaminated clothing before reuse.
- For hot product, immediately immerse in or wash the affected area with large amounts of cold water to dissipate heat.
- Get emergency medical treatment.
- If skin irritation or erythema occurs, seek medical treatment and advice.
- In case of minor skin contact, prevent spread of contaminated areas.
- Removal of solidified molten material from skin requires medical assistance.

○ If inhaled

- If you inhaled steam, immediately move to a place with fresh air.
- Give artificial respiration if victim is not breathing.
- If breathing is difficult, supply oxygen.
- Talk to your doctor.
- If exposed or concerned about exposure, seek medical treatment and advice.
- Keep victim warm and at rest
- Get emergency medical treatment.
- If breathing becomes difficult after inhalation, move to a place with fresh air and rest in a position that is easy for breathing.
- If respiratory symptoms occur, consult a medical institution (doctor).
- If exposed to excessive levels of dusts or fumes, remove with fresh air and get medical attention if cough or other symptoms develop.

○ If swallowed

- Rinse mouth with water.
- Warm and stabilize the victim.
- Do not give anything by mouth to an unconscious person.
- Talk to your doctor.
- If exposed or concerned about exposure, seek medical treatment and advice.
- Get emergency medical treatment.
- If you feel discomfort after swallowing, consult a medical institution (doctor).

○ Other medical attention.

- Ensure that medical personnel are aware of the material(s) involved and take precautions to protect themselves.
- In case of exposure, contact medical staff and take special emergency measures such as follow-up investigation.

○ Potential health effect

- Fatal if inhaled
- May cause an allergic skin reaction

5. FIRE-FIGHTING MEASURES

○ **Suitable extinguishing media**

- Suitable extinguishing media: alcohol foam, carbon dioxide or water spray, dry sand or soil for asphyxiation.
- Unsuitable extinguishing media: high-pressure water

○ **Specific hazards arising from the chemical**

- Fire may produce irritating and/or toxic gases.
- It can re-ignite after extinguishment.
- Some substances burn with intense heat.
- Inhalation and contact of vapors, substances, and degradation products may result in serious injury or death.
- Leaked material poses a fire/explosion risk.
- Some substances can flash and burn quickly.
- Unstable at room temperature
- May explode from heat, flame, friction or contamination.
- Containers may explode when heated.
- Flammable/combustible substances

○ **Special protective equipment for fire-fighters**

- Wear a respirator and safety gear before entering the fire area.
- Move containers from fire area if you can do it without risk.
- Wear positive pressure self-contained breathing apparatus (SCBA).
- Rescuers should put on appropriate protective gear.
- Evacuate area and fight fire from a safe distance.
- Do not get water inside containers.
- Be careful as it may melt and be transported.
- Dike fire-control water for later disposal; do not scatter the material.
- If extinguishing is impossible, protect surroundings and let the fire extinguish itself.
- Fire involving Tanks: Withdraw immediately in case of high-pitched sound from pressure relief devices or discoloration of tank.
- Fire involving Tanks: ALWAYS stay away from tanks engulfed in fire.

6. ACCIDENTAL RELEASE MEASURES

○ **Personal precautions**

- Avoid contact with eyes.
- ELIMINATE all ignition sources (no smoking, flares, sparks or flames in immediate area).
- Do not come into contact with or cross the spills.
- Please note materials and conditions to be avoided.
- Ventilate the contaminated area.
- Do not enter the space without adequate protective equipment, such as air respirators or air masks, until adequate air (oxygen concentration of 18-23.5%) is obtained.
- Do not inhale dust or fumes.
- Isolate contaminated area.
- Keep unnecessary and unprotected personnel from entering.
- Stop leak if it is not dangerous.
- Do not touch damaged containers or spilled material unless wearing appropriate protective clothing.
- Prevent the formation of dust.
- Since the fine particles can cause a fire or explosion, eliminate all ignition sources.
- Clean up spills immediately, observing precautions in Protective Equipment section.

○ **Environmental precautions**

- Prevent entry into waterways, sewers, basements or confined areas.
- Prevent entry into sewers or drains.
- Leakage may cause contamination.
- Do not discharge into the environment.

○ **Methods and materials for containment and cleaning up**

- In case of small leak, wash the contaminated area with plenty of water.
- In case of small leak, absorb with sand or non-combustible material and place in a container.
- Use a clean shovel to scoop the spill into a clean, dry container, close loosely and move the container away from the spill area.
- Dispose of and collect the materials preventing dust generation.
- Store in appropriate containers for disposal.
- Collect the leakages.

7. HANDLING AND STORAGE

○ **Precautions for safe handling**

- Avoid contact with eyes.
- When handling containers, it is recommended to use appropriate mechanical devices for safety.
- When injecting, avoid direct contact with skin and eyes. Wash thoroughly after handling.
- Be careful of fire caused by flames, sparks, etc.

- When working, wear appropriate personal protective equipment according to “Section 8”.
- Please note materials and conditions to be avoided.
- Do not handle until all safety precautions have been read and understood.
- Use with caution when handling/storing.
- Do not enter storage area without proper ventilation.
- Prevent dust generation.
- Avoid inhalation of dust or fumes.
- Wash the handling area thoroughly after handling.
- Do not eat, drink or smoke when using this product.
- Handle only outdoors or in a well-ventilated area.
- Do not take contaminated clothing out of the workplace.
- Avoid prolonged or continuous skin contact.
- Be careful of high temperatures.
- Wet activated carbon consumes oxygen in the air, so in confined spaces (closed spaces) the oxygen concentration can drop to dangerous levels.

○ **Conditions for safe storage**

- Avoid areas where packaging containers may be damaged or contaminated.
- Store in a dry place with good ventilation and away from direct sunlight or heat sources.
- Choose a place that can be protected from oxidants and acids.
- Keep away from heat, sparks, open flames and high temperatures - No smoking
- Since material can spontaneously ignite (pyrophoric) when exposed to room temperature or slightly elevated temperatures, store it at the proper temperature.
- Please note materials and conditions to be avoided.
- Keep away from food and drinking water.
- Empty drums should be completely drained, properly bunged, and promptly returned to a drum reconditioner, or properly disposed.
- Keep at low temperature and avoid direct sunlight.
- Store the container tightly closed in a well-ventilated place.
- Maintain spacing between cargoes.

8. EXPOSURE CONTROL/PERSONAL PROTECTION

○ **Components with workplace control parameter**

KOSHA :

Ingredients	TWA	STEL
Iron	1 mg/㎥	-

Copper	1 mg/㎥ (dust and mist), 0.1 mg/㎥ (fume)	2mg/㎥ (dust and mist)
Carbon	5mg/㎥ (activated carbon)	-

ACGIH :

Ingredients	TLV	STEL
Iron	-	-
Copper	0.2 mg/㎥ (Fume), 1 mg/㎥ (Dusts and mists)	-
Carbon	-	-

BEI : Not applicable

○ Appropriate engineering controls :

- Use process enclosures, local exhaust ventilation, or other engineering controls to control airborne levels below recommended exposure limits.
- Install the eyewash facility and a safety shower near the workshop.

○ Personal protective equipment

Respiratory protection :

- Use certified respiratory protection suitable for the physical and chemical properties of the particulate matter to which you are exposed.
- In case exposed to particulate material, the respiratory protective equipments as follow are recommended. ;facepiece filtering respirator or air-purifying respirator, high-efficiency particulate air(HEPA) filter media or respirator equipped with powered fan, filter media of use (dust, mist, fume)
- In lack of oxygen (< 19.6%), wear the supplied-air respirator or self-contained oxygen breathing apparatus.
- If the exposure concentration is lower than 10 mg/㎥, wear a respiratory protective gear of half mask type that has an appropriate type filter.
- If the exposure concentration is lower than 25 mg/㎥, wear a loose-fitting hood/helmet powered respirator or continuous flow dust mask equipped with an appropriate type of filter.
- If the exposure concentration is lower than 50 mg/㎥, wear a full-face or powered half-type or air-supplied continuous-flow/pressure-demand half-type respirator equipped with an appropriate filter.
- If the exposure concentration is lower than 1,000 mg/㎥, wear a full-face or helmet/hood type, pressure-demanded type air supply mask equipped with an appropriate filter.

- If the exposure concentration is lower than 10,000 mg/㎥, wear self-contained breathing apparatus (SCBA) or self-contained breathing apparatus operated in pressure demand mode with appropriate filter.

Eye protection :

- Wear safety glasses and face shield of appropriate material considering the physical and chemical properties of the chemical.

Hand protection :

- Wear protective gloves of appropriate material considering the physical and chemical properties of the chemical.

Skin and body protection :

- Wear protective gloves/protective clothing/safety glasses/face shield/earplugs of appropriate material considering the physical and chemical properties of the chemical.

9. PHYSICAL AND CHEMICAL PROPERTIES

○ **State** :

Appearance : Solid or Solid powder

Color : Grey, Black

○ **Odor** : Not available

○ **Odor Threshold** : Not available

○ **pH** : Not applicable

○ **Melting range (Initial)** : 1,538 ℃ (Iron) ※ from US NLM / ECHA

○ **Boiling point (Initial)** : 2,861 ℃ (Iron) ※ from US NLM / ECHA

○ **Flash point** : Not available

○ **Autoignition temperature** : Not available

○ **Evaporation rate** : Not available

○ **Flammability** : Combustible ※ from ECHA

○ **Lower explosion limit / Upper explosion limit** : Not available

○ **Combustion characteristics** : Not available

○ **Vapour pressure** : 1 ㎩ (1,455℃)

○ **Water solubility** :

Water solubility: insoluble.

Solvent solubility: solubility - acid, insoluble - alkali, alcohol, ether

○ **Vapor density:** Not available

○ **Relative density** : 8.9 (water = 1)

○ **Partition coefficient (n-octanol/water)** : Not available

○ **Pyrophoric temperature:** Not available

○ **Decomposition temperature** : Not available

○ **Viscosity** : Not available

○ **Molecular Weight** : 55.85

10. STABILITY AND REACTIVITY

○ **Chemical stability**

- Stable at normal temperature and pressure.
- Inhalation of material may be harmful.
- It ignites itself when exposed to air.
- Leaked material poses a fire/explosion risk.
- It can re-ignite after extinguishment.
- It may ignite on contact with moisture.
- Flammable/combustible material.
- Some substances can flash and burn quickly.
- Inhalation of degradation products may result in serious injury or death.
- Fire may produce irritating, corrosive and/or toxic gases.
- It may explode from heat, flame, friction or contamination.
- Some substances burn with intense heat.

○ **Conditions to avoid**

- Avoid formation of dust.
- Keep away from heat, sparks, open flames and high temperatures - No smoking
- Material can spontaneously ignite (pyrophoric) when exposed to room temperature or slightly elevated temperatures.
- Water (There is a possibility of producing corrosion).

○ **Materials to avoid**

- Strong acid, strong base.
- Combustible material, reducing substance.

○ **Hazardous decomposition products**

- Fire may produce irritating and/or toxic gases.
- Corrosive/toxic fume.

11. TOXICOLOGICAL INFORMATION

○ **Exposure route information.**

- Fatal if inhaled
- May cause an allergic skin reaction

○ **Acute toxicity**

Oral : Not classified (ATEmix = 103,958.7 mg/kg)

Substance name	Toxicity information	Reference
Iron	LD_{50} = 98,600 mg/kg ((rat), OECD Guideline 401) (Read across: hydrogen reduced iron powder)	※ from ECHA
Copper	LD_{50} > 2,500 mg/kg ((rat), OECD Guideline 423, GLP) (Read across: Copper oxide)	※ from ECHA
Carbon	LD_{50} > 2,000 mg/kg ((rat), OECD Guideline 423, GLP)	※ from ECHA

Dermal : Not classified

Substance name	Toxicity information	Reference
Iron	Not available	
Copper	LD_{50} > 2,000 mg/kg ((rat), OECD Guideline 402, GLP) (Read across: Coated copper flakes)	※ from ECHA
Carbon	Not available	

Inhalation : Cat. 2 (ATEmix = 0.4 mg/L / 4hr)

Substance name	Toxicity information	Reference
Iron	LC_{50} > 0.375 mg/L air / 4hr (rat) (Read across: carbonyl iron)	※ from ECHA
Copper	LC_{50} > 5.11 mg/L air / 4hr ((rat), OECD Guideline 436, GLP) (Read across: Coated copper flakes)	※ from ECHA
Carbon	LC_{50} > 2.125 mg/L air / 4hr ((rat), OECD Guideline 403)	※ from ECHA

○ **Skin irritation** : Not classified

Substance name	Toxicity information	Reference
Iron	Not irritating ((rabbit), OECD Guideline 404, GLP) (Read across:	※ from ECHA

	bayferrox-schwarz 350 fluessig)	
Copper	Not irritating, A single 4hr, semi-occluded application of the test material to the intact skin of three rabbits produced no evidence of skin irritation. ((rabbit), OECD Guideline 404, GLP) (Read across: Coated copper flakes)	※ from ECHA
Carbon	Not irritating, no erythema or oedema effects after the application of activated carbon within 72 hours ((rabbit), OECD Guideline 404, GLP)	※ from ECHA

○ **Eye irritation** : Not classified

Substance name	Toxicity information	Reference
Iron	Not irritating ((rabbit), OECD Guideline 405) (Read across: Bayferrox-Schwarz 350 fluessig)	※ from ECHA
Copper	Slightly irritating, a slight potential to induce eye irritation ((rabbit), OECD Guideline 405, GLP) (Read across: Coated copper flakes)	※ from ECHA
Carbon	Not irritating, cornea opacity score = 0, iris score = 0, conjunctivae score = 0.67, chemosis score = 0.22 ((rabbit), OECD Guideline 405, GLP)	※ from ECHA

○ **Respiratory sensitization** : Not available

○ **Skin sensitization** : Cat. 1

Substance name	Toxicity information	Reference
Iron	Not sensitizing (guinea pig) (Read across: 1309-37-1 etc)	※ from ECHA
Copper	It was classified in Category 1A because the Japan Society for Occupational Health (JSOH) classified copper and its compounds as occupational skin sensitizers Group 2 (Recommendation of Occupational Exposure Limits (Japan Society For Occupational Health (JSOH), 2012))	※ from NITE
Carbon	Not sensitizing (mouse), (OECD Guideline 429, GLP)	※ from ECHA

○ **Germ cell mutagenicity** : Not classified

Substance name	Toxicity information	Reference
Iron	In vitro - Positive (mouse lymphoma cells L5178Y, in vitro mammalian cell gene mutation assay, Metabolic activation: with and without. OECD Guideline 476) (Read across: electrolytic iron powder)	※ from ECHA
Copper	In vitro - Negative (Salmonella typhimurium TA98, TA100, TA1535, TA1537, TA102, bacterial reverse mutation assay, Metabolic activation: with and without, OECD Guideline 471, GLP) (Read across: 7758-99-8) In vivo - Negative (mouse), micronucleus assay, EU Method B.12, GLP) (Read across: 7758-99-8)	※ from ECHA
Carbon	In vitro - Negative (Salmonella typhimurium TA 1535, TA 1537, TA 98, TA 100, TA 102, bacterial reverse mutation assay, Metabolic activation: with and without, OECD Guideline 471, GLP)	※ from ECHA

○ **Carcinogenicity** : Not classified

Substance name	Toxicity information	Reference
Iron	MOEL Notice, IARC, ACGIH, NTP, EU CLP - Not applicable	※ from MOEL Notice, IARC, ACGIH, NTP, EU CLP
Copper	MOEL Notice, IARC, ACGIH, NTP, EU CLP - Not applicable	※ from MOEL Notice, IARC, ACGIH, NTP, EU CLP
Carbon	MOEL Notice, IARC, ACGIH, NTP, EU CLP - Not applicable	※ from MOEL Notice, IARC, ACGIH, NTP, EU CLP

○ **Reproductive toxicity** : Not classified

Substance name	Toxicity information	Reference
Iron	Not available	
Copper	No reproductive toxicity was seen at any concentration. ((rat), OECD Guideline 416, GLP) (Read across: 7758-99-8)	※ from ECHA
Carbon	Not available	

○ **Specific target organ toxicity - single exposure (GHS)** : Not classified

Substance name	Toxicity information	Reference
Iron	6 h exposure to 250 mg/m^3 respirable iron powder (carbonyl iron) does not result in mortality. (rat) (Read across: carbonyl iron)	※ from ECHA
Copper	Signs of systemic toxicity noted in animals treated with 2,000 mg/kg bw were hunched posture, lethargy, pilo-erection, diarrhoea, decreased respiration rate, laboured respiration, ataxia, pallor of the extremities, emaciation, tiptoe gait and faeces stained green. Hunched posture was noted during the day of dosing and one day after dosing in one male treated with 200 mg/kg bw. No other signs of systemic toxicity were noted in animals treated with 200 mg/kg bw. ((rat), OECD Guideline 423, GLP) (Read across: Coated copper flakes)	※ from ECHA
Carbon	Animals visible exhibited labored breathing and intermittent gasping. Upon removal from the chamber mucoid nasal discharge, salivation, rapid or labored breathing, gasping, dry or moist rales, reduced activity, and wet or matted ano-genital fur were observed ((rat), OECD Guideline 403)	※ from ECHA

○ **Specific target organ toxicity - repeated exposure (GHS)** : Not classified

Substance name	Toxicity information	Reference
Iron	In a subacute inhalation study with carbonyl iron, rats showed a clear inflammatory reaction in the lungs, as well as affected clearance, increased cell proliferation, hypertrophy and hyperplasia at 50 and 250 mg/m^3. (rat)	※ from ECHA

Copper	The overarching findings of this study were the exposure level-dependent appearance of macrophages in the lung, an increase in neutrophil number in BALF as well as in blood, and an increase in LDH and protein levels in the BALF. An increase in inflammation scores (neutrophil-dominated inflammation) was observed in the lung (the highest score being “mild”), and there was a decrease in the wet/dry lung weight ratio (highest exposure level only). Some nasal findings were reported for the high and medium-high exposures in the males. (rat), OECD Guideline 412, GLP) (Read across: Cu^{2+} as cuprous oxide)	※ from ECHA
Carbon	The very slight lung inflammatory changes based on minimal increases in pulmonary neutrophils seen at the 3.33 mg/m3 and 7.29 mg/m3 dose levels are considered responses within the normal physiological range. (rat), OECD Guideline 413, GLP)	※ from ECHA

○ **Aspiration hazard** : Not available

12. ECOLOGICAL INFORMATION

○ Toxicity

Hazardous to the aquatic environment – short-term (acute) aquatic hazard – Cat. 1

Hazardous to the aquatic environment – long-term (chronic) aquatic hazard – Cat. 2

(Chronic 1 ingredient content multiplied by a weight of 10 is 25% or more (Copper)

Fish: Not classified (ATEmix = 2.431 mg/L)

Substance name	Toxicity information	Reference
Iron	LC_{50}= 8.65 mg/L (96hr, Oncorhynchus mykiss)	※ from ECHA
Copper	LC_{50} = 0.164 mg/L (96hr, Oncorhynchus mykiss) / NOEC = 0.066 mg/L (270d, Pimephales promelas)	※ from ECHA
Carbon	LC_{50} = 165.536 mg/L (96hr)	※ from EPI SUITE

Crustacean: Not classified

Substance name	Toxicity information	Reference
Iron	EC_{50} > 100 mg/L (48hr, OECD Guideline 202, GLP, daphnia magna) (Read across: 1309-37-1)	※ from ECHA
Copper	EC_{50} = 0.0338 mg/L (48hr, daphnia magna) / NOEC = 0.3 mg/L (28d, Oncorhynchus mykiss)	※ from ECHA
Carbon	Not available	

Algae: Cat. 1 (ATEmix = 0.62 mg/L)

Substance name	Toxicity information	Reference
Iron	EC_{50}= 18 mg/L (72hr, OECD Guideline 201, Raphidocelis subcapitata)	※ from ECHA

Copper	EC_{50} = 0.032 ~ 0.245 mg/L (72hr, OECD Guideline 201, pseudokirchneriell a subcapitata) (Read across: Coated copper flakes)	※ from ECHA
Carbon	EC_{50} = 39.195 mg/L (96hr)	※ from EPI SUITE

○ Persistence and degradability :

Substance name	Toxicity information	Reference
Iron	log Kow = -0.77 / Ready Biodegradability Prediction: Yes	※ from EPI SUITE
Copper	log Kow = -0.57 / Ready Biodegradability Prediction: Yes	※ from ICSC / EPI SUITE
Carbon	log Kow = 0.78 / Ready Biodegradability Prediction: Yes	※ from EPI SUITE

○ Bioaccumulative potential :

Substance name	Toxicity information	Reference
Iron	BCF = 0.03 ~ 0.08 L/kg	※ from HSDB
Copper	BCF = 3.162 L/kg	※ from EPI SUITE
Carbon	BCF = 2.433 L/kg	※ from EPI SUITE

○ Mobility in soil :

Substance name	Toxicity information	Reference
Iron	Koc = 13.22	※ from EPI SUITE
Copper	Koc = 13.22	※ from EPI SUITE
Carbon	Koc = 13.22	※ from EPI SUITE

○ Other adverse effects

- Ozone layer : Not available

13. DISPOSAL CONSIDERATIONS

○ Disposal consideration

- Dispose of waste appropriately through a licensed industrial waste disposal company in accordance with the Waste Management Act and other related regulations.

○ Disposal precaution

- A business operator discharging business waste (business waste discharger) shall dispose of waste generated at a business site by himself/herself, or delegate it to a waste disposal business operator, a person who regenerates other people's waste, and a person who installs and operates a waste disposal facility.

- Dispose of contents/containers in accordance with waste-related laws and regulations.

14. TRANSPORT INFORMATION

○ UN RTDG :

UN Number : 1383

UN Proper shipping name : PYROPHORIC METAL, N.O.S. or PYROPHORIC ALLOY, N.O.S.

Transport Hazard class : 4.2

Packing group : I

Marine pollution : Not applicable

○ Special precaution

Fire EmS Guide : F-G

Spillage EmS Guide : S-M

15. REGULATORY INFORMATION

○ Korea Industrial Safety and Health Act :

Regulatory criteria	Substance name
Hazardous Substances Subject to Control	Iron, Copper
Special management materials	Not applicable
Harmful Agents Subject to Work Environment Monitoring	Copper(6months)
Harmful Agents Subject to Workers Requiring Health Examination	Copper(12months)
Threshold Limit Values (TLVs) chemicals	Iron, Carbon, Copper
Chemicals subject to permissible exposure limit	Not applicable
Matters subject to Submission of Process Safety Reports	Not applicable
Prohibited Substance	Not applicable
Substance subject to authorization	Not applicable

○ Korea Chemicals Control Act : Not applicable

○ Korea Hazardous Materials Safety Control Act :

Substance name	Substance of regulation.
Iron	Iron content class 2, 500kg
Copper	Non- Hazardous Substances

Carbon	Non- Hazardous Substances

○ Wastes Control Act:

- This product is industrial waste which is not the designated waste defined in Table 1 of the Enforcement Decree of the Wastes Control Act.
- Designated waste: Iron, Copper, Carbon

○ Regulations under other domestic and foreign laws:

Act on the Registration and evaluation, etc. of Chemical Substances

Substance name	Substance of regulation.
Iron	Phase-in substance (KE-21059)
Copper	Phase-in substance (KE-08896)
Carbon	Phase-in substance (KE-04671)

Korea Persistant Organic Pollutants Control Act : Not applicable

EU Classification :

Substance name	Classification result	Hazard statements
Iron	Not classified	-
Copper	Aquatic Chronic 2 (granulated copper; [particle length: from 0,9 mm to 6,0 mm; particle width: from 0,494 to 0,949 mm])	H411
Carbon	Not classified	–

16. OTHER INFORMATION

○ **Issued Date** : 2017. 11. 01

○ **Revision No.** : 5

○ **Revision Date** : 2023. 08. 30

○ **References**

- ECHA : European chemical agency, http://echa.europa.eu/
- US NLM : U.S. National Library of Medicine, http://chem.sis.nlm.nih.gov/chemidplus/
- HSDB : U.S. Hazardous Substances Data Bank, http://toxnet.nlm.nih.gov/
- CCRIS : U.S. Chemical Carcinogenesis Research Information System, http://toxnet.nlm.nih.gov/
- IARC : International Agency for Research on Cancer, http://monographs.iarc.fr/

- JP NITE : Japan National Institute of Technology and Evaluation, http://www.safe.nite.go.jp/
- KOSHA: Korea Occupational Safety and Health Agency, https://msds.kosha.or.kr/MSDSInfo/
- KFI : Korea Fire Institute, https://hazmat.mpss.kfi.or.kr/index.do
- NCIS : National Chemicals Information System, https://ncis.nier.go.kr/main.do
- EPI Suite : Estimation Programs Interface Suite,
https://www.epa.gov/tsca-screening-tools/epi-suitetm-estimation-program-interface
- GESTIS, https://gestis-database.dguv.de/search
- PubChem, https://pubchem.ncbi.nlm.nih.gov/
- UN RTDG : Recommendations on the transport of dangerous goods

○ **Inquiry**

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